Log Linearization

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Business cycle is fluctuation / deviation around trend. Log linearization: take logs to linearize the system and then take deviation from the trend / steady state.

* Can not work with discontinuous functions, eg: zero lower bound, default, work choice,etc.

1 Taylor expansions

Taylor expansion of order k of function f around a is to derive the best approximation of f around a by a polynomial of order k.

Suppose f is $n+1$ times differentiable in an interval around a

$$x \in (a-\varepsilon,\ a+\varepsilon),$$

then

$$f(x) = f(a) + \sum_{i=1}^{n} \frac{f^{(i)}(a)}{i!}(x-a)^i + \underbrace{o\left((x-a)^n\right)}_{\text{Peano remainder terms.}}.$$

eg:

$$\text{1st order:} \qquad f(x) \approx f(a) + \frac{f'(a)}{1!}(x-a)^1 = f(a) + f'(a)(x-a)$$

$$\text{2nd order:} \qquad f(x) \approx f(a) + \frac{f'(a)}{1!}(x-a)^1 + \frac{f''(a)}{2!}(x-a)^2 = f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2$$

2 Taylor expansion on two dimensions

Suppose f is second-order continuous (C^2) in a circle around (x_0, y_0) that contains $(x_0 + h, y_0 + k)$.

For a small move (h, k) around (x_0, y_0), how to approximate changes in f.

$$f(x_0+h, y_0+k) = f(x_0, y_0) + \Big[f_1(x_0, y_0)h + f_2(x_0, y_0)k\Big] + \frac{1}{2}\Big[f_{11}(x, y)h^2 + 2f_{12}(x, y)hk + f_{22}(x, y)k^2\Big]$$

where

$$(x, y) = (x_0 + ch, y_0 + ck) \qquad \text{for some } c \in (0, 1).$$

A vector format:

$$f(z_0 + \Delta) = f(z_0) + \nabla f(z_0)^T \Delta + \frac{1}{2}\Delta^T H_f(z_0 + c\Delta)\Delta$$

$$\text{where } z_0 = (x_0, y_0), \qquad \Delta = (h, k)$$

3 Log linearization

$$\text{percentage deviation:} \qquad \hat{x}_t = \log X_t - \underbrace{\log \bar{X}}_{Steadystate}$$

Since $Y_t \to 0$ when Y_t is sufficiently small, we have

$$e^{Y_t} = e^0 + e^0(Y_t - 0) = 1 + Y_t \qquad \Rightarrow \qquad Y_t \approx \log(1 + Y_t).$$

$$\hat{x}_t = \log \frac{X_t}{\bar{X}} = \log\left(1 + \frac{X_t - \bar{X}}{\bar{X}}\right) \approx \frac{X_t - \bar{X}}{\bar{X}}.$$

$\hat{x}_t \times 100\%$ is approximately the percentage deviation w.r.t. S.S.

$$X_t = \bar{X}\left(\frac{X_t}{\bar{X}}\right) = \bar{X}e^{\log(X_t/\bar{X})} = \bar{X}e^{\hat{x}_t}$$

Taking first-order Taylor approximation:

$$\bar{X}e^{\hat{x}_t} \approx \bar{X}e^0 + \bar{X}e^0(\hat{x}_t - 0) = \bar{X} + \bar{X}\hat{x}_t = (1 + \hat{x}_t)\bar{X}$$

$$\Rightarrow \quad X_t = \overbrace{\bar{X}}^{\text{Steady state}} + \overbrace{\bar{X}\hat{x}_t}^{\text{deviation from steady state}}$$

4 Examples

4.1 Linear equation

$$Y_t = aX_t$$

$$X_t \approx \bar{X}(1 + \hat{x}_t) \quad \Rightarrow \quad Y_t \approx a\bar{X}(1 + \hat{x}_t) = a\bar{X} + a\bar{X}\hat{x}_t$$

Since

$$Y_t = aX_t,\ \forall t \quad \Rightarrow \quad \bar{Y} = a\bar{X},$$

we get

$$Y_t \approx \bar{Y} + \bar{Y}\hat{x}_t \quad \Rightarrow \quad Y_t \approx \bar{Y} + \bar{Y}\hat{y}_t$$

hence

$$\hat{x}_t = \hat{y}_t.$$

4.2 Product

$$Z_t = X_t Y_t$$

$$X_t \approx \bar{X}(1 + \hat{x}_t), \qquad Y_t \approx \bar{Y}(1 + \hat{y}_t)$$

$$\Rightarrow \quad Z_t \approx \bar{X}\bar{Y}(1 + \hat{x}_t)(1 + \hat{y}_t) = \bar{X}\bar{Y}\left(1 + \hat{x}_t + \hat{y}_t + \underbrace{\hat{x}_t\hat{y}_t}_{\approx 0}\right)$$

So

$$Z_t \approx \bar{X}\bar{Y}(1 + \hat{x}_t + \hat{y}_t).$$

Since

$$Z_t = X_t Y_t,\ \forall t \quad \Rightarrow \quad \bar{Z} = \bar{X}\bar{Y},$$

we get

$$Z_t \approx \bar{Z} + \bar{Z}(\hat{x}_t + \hat{y}_t) \quad \Rightarrow \quad Z_t \approx \bar{Z} + \bar{Z}\hat{z}_t$$

hence

$$\hat{z}_t = \hat{x}_t + \hat{y}_t.$$

4.3 Constant product

Log linearize equation

$$X_t Y_t = C \qquad \Rightarrow \qquad \bar{X}\bar{Y} = C.$$

By (2),

$$X_t Y_t \approx \bar{X}\bar{Y}(1 + \hat{x}_t + \hat{y}_t) \qquad \Rightarrow \qquad C \approx C(1 + \hat{x}_t + \hat{y}_t)$$

so

$$\hat{x}_t + \hat{y}_t = 0.$$

4.4 Ratio

$$\frac{X_t}{Y_t} = C \qquad \Rightarrow \qquad \frac{\bar{X}}{\bar{Y}} = C.$$

$$\frac{X_t}{Y_t} \approx \frac{\bar{X}(1 + \hat{x}_t)}{\bar{Y}(1 + \hat{y}_t)}$$

Use

$$(1 + \hat{y}_t)(1 - \hat{y}_t) = 1 + \hat{y}_t - \hat{y}_t - \hat{y}_t\hat{y}_t \qquad \Rightarrow \qquad \frac{1}{1 + \hat{y}_t} \approx (1 - \hat{y}_t).$$

Therefore

$$\frac{X_t}{Y_t} = \frac{\bar{X}}{\bar{Y}}(1 + \hat{x}_t)(1 - \hat{y}_t) \approx \frac{\bar{X}}{\bar{Y}}\left(1 + \hat{x}_t - \hat{y}_t - \hat{x}_t\hat{y}_t\right)$$

So

$$\frac{X_t}{Y_t} \approx \frac{\bar{X}}{\bar{Y}}(1 + \hat{x}_t - \hat{y}_t).$$

Since

$$\frac{X_t}{Y_t} = \frac{\bar{X}}{\bar{Y}} = C,$$

we have

$$C = C(1 + \hat{x}_t - \hat{y}_t) \qquad \Rightarrow \qquad \hat{x}_t - \hat{y}_t = 0.$$

4.5 Sum

$$X_t + Y_t = C \qquad \Rightarrow \qquad \bar{X} + \bar{Y} = C.$$

$$X_t + Y_t \approx \bar{X}(1 + \hat{x}_t) + \bar{Y}(1 + \hat{y}_t) = \bar{X} + \bar{Y} + \bar{X}\hat{x}_t + \bar{Y}\hat{y}_t$$

hence

$$\bar{X}\hat{x}_t + \bar{Y}\hat{y}_t = 0.$$

4.6 Budget constraint

$$C_t + K_t = Y_t + (1-\delta)K_{t-1} \qquad \Rightarrow \qquad \bar{C} + \bar{K} = \bar{Y} + (1-\delta)\bar{K}$$

$$C_t \approx \bar{C}(1+\hat{c}_t), \qquad K_t \approx \bar{K}(1+\hat{k}_t), \qquad Y_t \approx \bar{Y}(1+\hat{y}_t), \qquad K_{t-1} \approx \bar{K}(1+\hat{k}_{t-1})$$

$$\Rightarrow \quad \bar{C}(1+\hat{c}_t) + \bar{K}(1+\hat{k}_t) = \bar{Y}(1+\hat{y}_t) + (1-\delta)\bar{K}(1+\hat{k}_{t-1})$$

$$\Rightarrow \quad \bar{C} + \bar{C}\hat{c}_t + \bar{K} + \bar{K}\hat{k}_t = \bar{Y} + \bar{Y}\hat{y}_t + (1-\delta)\bar{K} + (1-\delta)\bar{K}\hat{k}_{t-1}$$

Using

$$\bar{C} + \bar{K} = \bar{Y} + (1-\delta)\bar{K},$$

we obtain

$$\bar{C}\hat{c}_t + \bar{K}\hat{k}_t = \bar{Y}\hat{y}_t + (1-\delta)\bar{K}\hat{k}_{t-1}.$$

4.7 Utility FOC

$$C_t^{-\sigma} = \lambda_t \qquad \Rightarrow \qquad \bar{C}^{-\sigma} = \bar{\lambda}$$

Let

$$C_t = \bar{C}e^{\hat{c}_t}, \qquad \lambda_t = \bar{\lambda}e^{\hat{\lambda}_t}.$$

Then

$$C_t^{-\sigma} = \bar{C}^{-\sigma}e^{-\sigma\hat{c}_t} = \bar{\lambda}e^{\hat{\lambda}_t}.$$

Since $\bar{C}^{-\sigma} = \bar{\lambda}$,

$$e^{-\sigma\hat{c}_t} = e^{\hat{\lambda}_t}.$$

Using

$$e^{-\sigma\hat{c}_t} \approx 1 - \sigma\hat{c}_t, \qquad e^{\hat{\lambda}_t} \approx 1 + \hat{\lambda}_t,$$

we get

$$1 - \sigma\hat{c}_t = 1 + \hat{\lambda}_t \qquad \Rightarrow \qquad -\sigma\hat{c}_t = \hat{\lambda}_t.$$

4.8 C-D production function

$$Y_t = A_t K_t^{\alpha} N_t^{1-\alpha} \qquad \Rightarrow \qquad \bar{Y} = \bar{A}\bar{K}^{\alpha}\bar{N}^{1-\alpha}$$

$$\log Y_t = \log A_t + \alpha \log K_t + (1-\alpha)\log N_t$$
$$\log \bar{Y} = \log \bar{A} + \alpha \log \bar{K} + (1-\alpha)\log \bar{N}$$

Subtract:

$$\log Y_t - \log \bar{Y} = (\log A_t - \log \bar{A}) + \alpha(\log K_t - \log \bar{K}) + (1-\alpha)(\log N_t - \log \bar{N})$$

hence

$$\hat{y}_t = \hat{a}_t + \alpha\hat{k}_t + (1-\alpha)\hat{n}_t.$$

4.9 GDP identity

$$Y_t = C_t + I_t + G_t$$

From (5) we know

$$\hat{y}_t = \frac{\bar{C}}{\bar{Y}}\hat{c}_t + \frac{\bar{I}}{\bar{Y}}\hat{i}_t + \frac{\bar{G}}{\bar{Y}}\hat{g}_t = s_c\hat{c}_t + s_i\hat{i}_t + s_g\hat{g}_t.$$

share of GDP for each component

4.10 Capital law of motion

$$K_{t+1} = (1-\delta)K_t + I_t \qquad \Rightarrow \qquad \bar{K} = (1-\delta)\bar{K} + \bar{I}$$

$$\Rightarrow \quad \bar{K}(1+\hat{k}_{t+1}) = (1-\delta)\bar{K}(1+\hat{k}_t) + \bar{I}(1+\hat{i}_t)$$

$$\Rightarrow \quad \bar{K} + \bar{K}\hat{k}_{t+1} = (1-\delta)\bar{K} + (1-\delta)\bar{K}\hat{k}_t + \bar{I} + \bar{I}\hat{i}_t$$

Using

$$\bar{K} = (1-\delta)\bar{K} + \bar{I},$$

we get

$$\bar{K}\hat{k}_{t+1} = (1-\delta)\bar{K}\hat{k}_t + \bar{I}\hat{i}_t.$$

Thus

$$\hat{k}_{t+1} = (1-\delta)\hat{k}_t + \frac{\bar{I}}{\bar{K}}\hat{i}_t.$$

Since

$$\bar{K} = (1-\delta)\bar{K} + \bar{I} \qquad \Rightarrow \qquad \frac{\bar{I}}{\bar{K}} = \delta,$$

we obtain

$$\hat{k}_{t+1} = (1-\delta)\hat{k}_t + \delta\hat{i}_t.$$

4.11 Euler equation

$$U'(C_t) = \beta E_t\left[U'(C_{t+1})R_{t+1}\right]$$

use CRRA utility:

$$U(C_t) = \frac{C_t^{1-\sigma}}{1-\sigma} \qquad \Rightarrow \qquad U'(C_t) = C_t^{-\sigma}$$

So

$$C_t^{-\sigma} = \beta E_t\left[C_{t+1}^{-\sigma}R_{t+1}\right].$$

At steady state:

$$\bar{C}^{-\sigma} = \beta\bar{C}^{-\sigma}\bar{R} \qquad \Rightarrow \qquad 1 = \beta\bar{R}.$$

Let

$$C_t = \bar{C}e^{\hat{c}_t}, \qquad C_{t+1} = \bar{C}e^{\hat{c}_{t+1}}, \qquad R_{t+1} = \bar{R}e^{\hat{r}_{t+1}}.$$

Then

$$\bar{C}^{-\sigma}e^{-\sigma\hat{c}_t} = \beta E_t\left[\bar{C}^{-\sigma}e^{-\sigma\hat{c}_{t+1}}\bar{R}e^{\hat{r}_{t+1}}\right].$$

First-order approximation:

$$\bar{C}^{-\sigma}(1-\sigma\hat{c}_t) = \beta E_t\left[\bar{C}^{-\sigma}\bar{R}(1-\sigma\hat{c}_{t+1})(1+\hat{r}_{t+1})\right]$$

$$= \beta E_t\left[\bar{C}^{-\sigma}\bar{R}\left(1 + \hat{r}_{t+1} - \sigma\hat{c}_{t+1} - \underbrace{\sigma\hat{c}_{t+1}\hat{r}_{t+1}}_{\approx 0}\right)\right]$$

Using $1 = \beta\bar{R}$:

$$1 - \sigma\hat{c}_t = 1 + E_t(-\sigma\hat{c}_{t+1} + \hat{r}_{t+1}).$$

Hence

$$-\sigma\hat{c}_t = E_t(-\sigma\hat{c}_{t+1} + \hat{r}_{t+1})$$

so

$$\hat{c}_t = E_t(\hat{c}_{t+1}) - \frac{1}{\sigma}E_t(\hat{r}_{t+1}).$$

5 Log linearization cheat sheet for economics

(1) $XY: \quad \hat{x} + \hat{y}$ (2) $X/Y: \quad \hat{x} - \hat{y}$

(3) $X^\alpha: \quad \alpha\hat{x}$ (4) $\prod_i X_i^{\alpha_i}: \quad \sum_i \alpha_i \hat{x}_i$

(5) $\sum_i X_i: \quad \sum_i \frac{\bar{X}_i}{\sum_j \bar{X}_j}\hat{x}_i$ (6) $K' = (1-\delta)K + I: \quad \hat{k}' = (1-\delta)\hat{k} + \delta\hat{i}$

(7) $\pi_t = \frac{P_t}{P_{t-1}}: \quad \hat{\pi}_t = \hat{p}_t - \hat{p}_{t-1}$ (8) CRRA MU : $\widehat{u'(c)} = -\sigma\hat{c}$

(9) CES index $X = \left[\sum_i w_i X_i^\rho\right]^{1/\rho} \quad : \quad \hat{x} = \sum_i s_i \hat{x}_i$

with $s_i = \frac{w_i \bar{X}_i^\rho}{\sum_j w_j \bar{X}_j^\rho}$